

Objectives of System Testing

The main objective of system testing for the **Bundoora Family Clinic Management System** is to make sure that the complete system works correctly, reliably, and securely before it is ready for final implementation. System testing is important because the project contains several connected components, including the user interface, login system, different user roles, appointment-related functions, and database operations. Rather than testing each component separately, system testing focuses on checking how the entire application performs when all components are working together.

One of the major objectives is to verify that the system meets the **functional requirements** identified during the system development process. Important functions such as user registration, login, password recovery, appointment management, patient services, notifications, inventory-related functions, and administrative activities should operate as expected. The testing process should also confirm that buttons, links, forms, menus, and other navigation features take users to the correct pages and perform their intended actions.

Another important objective is to test **user authentication and role-based access**. The Bundoora Family Clinic system includes different types of users, such as administrators, doctors, patients, and pharmacists. System testing should confirm that users can successfully access the appropriate areas of the application and that restricted functions are not available to unauthorised users. This is particularly important in a healthcare-related system because patient and clinic information needs to be handled carefully.

System testing also aims to check the **accuracy of data input and processing**. Forms should accept appropriate information while identifying missing, incomplete, or incorrectly formatted input. Information submitted through the frontend should also be processed correctly by the backend and stored accurately in the database. Database operations such as creating, retrieving, updating, and managing records should work consistently without causing unnecessary duplication or loss of information.

A further objective is to evaluate the **usability and consistency of the user interface**. Users should be able to understand the system without unnecessary confusion. Page layouts, headings, forms, navigation menus, dashboards, buttons, and messages should remain clear and consistent throughout the application. The system should also provide meaningful validation or error messages when users make mistakes so that they understand what needs to be corrected.

The testing process also considers **security, reliability, compatibility, and responsiveness**. Login and access-control functions should help prevent unauthorised access to protected areas of the application. The system should perform consistently during normal use and should respond appropriately when invalid information is entered. The web interface should also remain usable across common browsers and different screen sizes so that users can access the clinic system conveniently.

Finally, system testing is intended to **identify defects before the final deployment of the system**. Problems such as broken links, incorrect navigation, form validation issues, authentication problems, database errors, inconsistent interfaces, and integration failures can negatively affect the user experience if they are not discovered early. By identifying and correcting these issues, the development team can improve the overall quality of the Bundoora Family Clinic Management System. Therefore, the overall objective of system testing is to confirm that the completed application satisfies its specified requirements and is stable, secure, user-friendly, and ready for final evaluation and implementation.